



Universidad Nacional Autónoma de México
Facultad de Ingeniería



Tarea 4

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Administración de Servicios de Internet

Grupo: 3

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Tarea 4

C1. Crea un clúster

Task 2. Submit a job

Score: 50 / 100

24m left

Lab setup

Your cluster will build for a couple of minutes.

Waiting for cluster creation operation...done.

Created [... example-cluster]

When you see a "Created" message, you're ready to move on.

Test completed task

Click **Check my progress** to verify your performed task. If you have successfully created a Managed Service for Apache Spark cluster, you will see an assessment score.

✓

Create a cluster

Check my progress

Assessment Completed!

My account

Andres Silverio Martinez

Member since 2024

24 credits

Google Cloud Career Launchpad - LatAm Member

Dashboard

Progress

Settings

Sign Out

Privacy

Terms

Project ID

qwiklabs-gcp-04-25272dace4d

Browser setup

C2. Calculo de PI

Open Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. [Learn more](#)

Don't show again

Dismiss

```
Job [410e610c289647bcb5a7daeff351f4fb] submitted.
Waiting for job output...
26/05/24 06:02:45 INFO SparkEnv: Registering MapOutputTracker
26/05/24 06:02:45 INFO SparkEnv: Registering BlockManagerMaster
26/05/24 06:02:45 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/05/24 06:02:45 INFO SparkEnv: Registering OutputCommitCoordinator
26/05/24 06:02:46 INFO MetricsSystemImpl: Loaded properties from hadoop-metrics2.properties
26/05/24 06:02:46 INFO MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
26/05/24 06:02:46 INFO MetricsSystemImpl: google-hadoop-file-system metrics system started
26/05/24 06:02:47 INFO DataprocSparkPlugin: Registered 188 driver metrics
26/05/24 06:02:47 INFO DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at example-cluster-m.us-central1-a.c.qwiklabs-gcp-04-25272dace4d9.internal./10.128.0.2:8032
26/05/24 06:02:47 INFO AHSPProxy: Connecting to Application History server at example-cluster-m.us-central1-a.c.qwiklabs-gcp-04-25272dace4d9.internal./10.128.0.2:10200
26/05/24 06:02:48 INFO Configuration: resource-types.xml not found
26/05/24 06:02:48 INFO ResourceUtils: Unable to find 'resource-types.xml'.
26/05/24 06:02:49 INFO YarnClientImpl: Submitted application application_1779602330957_0002
26/05/24 06:02:50 INFO DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at example-cluster-m.us-central1-a.c.qwiklabs-gcp-04-25272dace4d9.internal./10.128.0.2:8030
26/05/24 06:02:52 INFO GoogleCloudStorageImpl: Ignoring exception of type GoogleJsonResponseException; verified object already exists with desired state.
26/05/24 06:02:52 INFO GoogleHadoopOutputStream: hflush(): No-op due to rate limit (RateLimiter[stableRate=0.2qps]): readers will *not* yet see flushed data for gs://dataproc-temp-us-central1-635456032265-fcdhuurm/c91d4d72-7b48-4e95-9849-c9c32323965e/spark-job-history/application_1779602330957_0002.inprogress [CONTEXT ratelimit_period="1 MINUTES" ]
Pi is roughly 3.141209311412093
26/05/24 06:03:09 INFO DataprocSparkPlugin: Shutting down driver plugin. metrics=[action_http_patch_request=0, files_created=1, gcs_api_server_timeout_count=0, op_get_list_status_result_size=0, op_open=0, action_http_delete_request=2, gcs_api_time=2048, gcs_backoff_count=0, gcs_api_client_unauthorized_response_count=0, stream_read_close_operations=0, stream_read_bytes_backwards_on_seek=0, gs_filesystem_create=3, exception_count=0, gcs_exception_count=0, gcs_api_total_request_count=33, op_create=1, stream_read_vectorized_operations=0, gcs_metadata_request=17, gcs_api_client_bad_request_count=0, action_http_put_request=3, op_create_non_recursive=0, gcs_api_client_gone_response_count=0, gs_filesystem_initialize=2, stream_read_vectorized_incoming_ranges=0, stream_write_operations=0, gcs_list_dir_request=0, stream_read_operations=
```

C3. Sube un trabajo

What do you want to learn today?

Task 2. Submit a job

Score: 50 / 100 22m left

Lab setup

```
...
Pi is roughly 3.14118528
...
state: FINISHED
```

Test completed task

Click **Check my progress** to verify your performed task. If you have successfully submitted a job, you will see an assessment score.

Submit a job

Check my progress

Assessment Completed!

Andres Silverio Martinez

Member since 2024

24 credits

Google Cloud Career Launchpad - LatAm Member

Dashboard

Progress

Settings

Sign Out

Privacy Terms

Project ID

qwiklabs-gcp-04-25272dace4d

C4. Modificación de los nodos trabajadores

```
CLOUD SHELL
Terminal (qwiklabs-gcp-04-25272dace4d9) x + v Open Editor

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

student_02_5974fdd50f32@cloudshell:~ (qwiklabs-gcp-04-25272dace4d9) $ gcloud dataproc clusters update example-cluster --num-workers 4
Waiting on operation [projects/qwiklabs-gcp-04-25272dace4d9/regions/us-centrall/operations/eebe2baf-485d-33fb-9396-0f717618d500].
Waiting for cluster update operation...done.
Updated [https://dataproc.googleapis.com/v1/projects/qwiklabs-gcp-04-25272dace4d9/regions/us-centrall/clusters/example-cluster].
student_02_5974fdd50f32@cloudshell:~ (qwiklabs-gcp-04-25272dace4d9) $ gcloud dataproc clusters update example-cluster --num-workers 2
Waiting on operation [projects/qwiklabs-gcp-04-25272dace4d9/regions/us-centrall/operations/9065309f-b8b8-30dd-b660-f051alf6dd78].
Waiting for cluster update operation...done.
Updated [https://dataproc.googleapis.com/v1/projects/qwiklabs-gcp-04-25272dace4d9/regions/us-centrall/clusters/example-cluster].
student_02_5974fdd50f32@cloudshell:~ (qwiklabs-gcp-04-25272dace4d9) $
```

C5. ¿Los clúster pueden ser creados y escalados de manera rápida con una variedad de tipos de maquinas virtuales, tamaños de discos y número de nodos?

The screenshot displays a Google Cloud Skills Boost lab interface. At the top, there is a search bar with the text "What do you want to learn today?". Below this, a navigation bar shows "Task 4. Test your understanding" with a dropdown arrow, a score of "Score: 100 / 100", and a timer "13m left". A "Lab setup" button is also visible. The main content area is titled "Task 4. Test your understanding" and includes the instruction: "Below are multiple-choice questions to reinforce your understanding of this lab's concepts. Answer them to the best of your abilities." A question is presented: "Clusters can be created and scaled quickly with a variety of virtual machine types, disk sizes, and number of nodes." The answer options are "True" (selected with a green checkmark) and "False". On the right side, a user profile for "Andres Silverio Martínez" is shown, indicating he is a "Member since 2024" with "24 credits". Below the profile, it says "Google Cloud Career Launchpad - LatAm Member". A sidebar menu includes links for "Dashboard", "Progress", "Settings", and "Sign Out", along with "Privacy" and "Terms" at the bottom.

Referencias

- [1] Brito Segura, A. (2026). Arquitectura para Ciencia de Datos. Tema 8. [Documento de clase]. Facultad de Ingeniería, Universidad Nacional Autónoma de México.
- [2] Managed Service for Apache Spark: Qwik Start - Command Line. Recuperado el 24 de mayo de 2026. dirección: https://www.skills.google/focuses/585?catalog_rank=%7B%22rank%22%3A1%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&parent=catalog&search_id=83334065